AI to Advance U.S. Competence in Agriculture, Sciences, and Engineering

Response to Notice of Request for Information (RFI) on the Development of an Artificial Intelligence (AI) Action Plan

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Washington State University (WSU), the land-grant university for the State of Washington and an R1 research institution under the Carnegie classification system with world-renowned programs in agriculture, computing, engineering, health, and sciences, appreciates the opportunity to provide a response with actionable recommendations the Office of Science and Technology Policy (OSTP)'s 2025 Request for Information (RFI) on the development of Artificial Intelligence (AI) action plan.

I. Opportunity

Much like the Manhattan era of the '40s and Apollo moon mission era of the '60s, we are at a pivotal moment in the nation's history. AI provides a potential breakthrough opportunity to significantly transform the state of science, engineering, and health in the U.S. society. More specifically, a data revolution – fueled by a rapid and pervasive adoption of sensing, mobile, and advanced measurement technologies – has allowed science, engineering, and life science domains including health and agriculture to gather data at breakneck speed. This digital data explosion represents a treasure trove of information, opening the doors for scientific inquiry, discovery, and progress, at an extreme scale. Fortunately, U.S. is also leading the AI revolution in the world, making significant and foundational strides in AI technology, its research and development, and its practice at industrial scale. Therefore, the U.S. is at a key strategic advantage to capitalize on both these advances, through active integration of AI into the scientific enterprise of discovery, engineering technological innovation, revolutionizing agricultural food production, and delivering on the promise of human health and well-being. The realization of this strategic mission for AI to effect social good hinges on our ability to bring about a system-wide integration and incorporation of AI into every aspect of the commercial and public enterprise. This systemic assimilation holds the key to enabling a sustained pathway to economic prosperity, increased exports and food security, and fortified U.S. leadership in the global stage in the areas of science, technology, and innovation.

Selected examples of key opportunities where AI plays a major role in science and technology solutions include:

- AI for accelerated scientific discovery at all scales, from molecular to individual to population levels;
- AI for sustained intensification of agricultural production, breeding resilient crops and reducing crop loss, reducing labor- and economic- uncertainties, and increasing farm productivity and farm worker productivity;
- AI for optimized use and smart stewardship of environmental resources including water, energy, and soil, and for improved air quality;
- AI for securing our power grid, renewable sources, aviation fuels, and bioenergy alternatives;
- AI for accelerated drug discovery, early detection of diseases, preventing and mitigating the threats of infectious diseases, and improving human health and well-being;

- AI for advanced manufacturing, advanced semiconductor design, and advanced materials design; and
- AI for transportation, construction, and advanced structural design.

Academic institutions and in particular, public land-grant universities such as WSU will have to play a lead role in steering this AI revolution. First, these **academic institutions are the innovation engines of cutting-edge research in AI** and all other related technologies, as well as the main drivers of fundamental research in science, engineering, agriculture, and health domains. Therefore, these institutions are ideally positioned to serve as the breeding grounds of foundational AI research alongside its applications through **"use-inspired AI"** research in various diverse domains and at the interfaces between various disciplinary boundaries. Secondly, these i**nstitutions are also the primary engines of training the next-generation workforce development**. Students obtaining four-year degrees and advanced graduate-level degrees represent the future of our industry and research workforce, and their early exposure to AI and its uses will be key in securing the future of our AI industry as well as commercial sectors that would benefit from the use of AI. Finally, **academic institutions also serve as the primary hub for the broader community outreach** through agricultural extensions, industry training programs such as bootcamps, research and development partnerships with national labs and industry, and student camps and teacher-training at the K-12 level. These programs offer a vehicle to educate, train, and raise awareness of AI among the communities (including both urban and rural communities) across the socio-technological spectrum, thereby creating a sustained pathway and system in place to gradually introduce and increase the responsible use of AI and related tools into various private and public sectors of the society.

II. Recommendations

1) Investment in use-inspired AI research, development, and deployment

An increased investment is needed in use-inspired AI, where the research, development, and deployment of AI tools are all inspired by concrete uses. AI has numerous potential applications across various domains. Yet, the type and class of AI needed across these domains varies widely to suit their respective uses:

- **AI for science** involves accelerating discovery through smarter experimental design, reduction in candidates for testing through virtual screening, active and adaptive learning, automated hypothesis extraction toward causal inference, knowledge and rule extraction from high dimensional and relational data, reasoning frameworks for process modeling, foundation models for scientific exploration and to create simulation abstractions.
- **AI for engineering** involves optimization, design space exploration and design automation, stress and extreme-condition testing, robotics and automation, transferability across platforms and systems and transferability from simulation to real-world, reproducibility and simulation capabilities through digital twins.
- **AI for agriculture** involves precision agriculture, trait optimization, robotics, automation and human-AI workflows to reduce labor costs and improve farm productivity, site-specific models for decision support, spatio-temporal modeling, robust training under uncertainty and using limited and noisy data, predictive and forecasting tools, crop response modeling, soil health, water security and water allocation, digital twins for plants, farms, and regions, earth-systems and biophysical modeling of complex systems, accelerated breeding of resilient and resistant varieties, community and economic modeling for adoption, interactive dashboards for decision making, and responsible and safe AI.
- **AI for health** involves drug design, early disease detection and prevention science through high-throughput screening and imaging, precision medicine, variant detection at molecular, cellular, tissue-specific, and organismal scales, understanding the roles of genetic and environmental variability on diseases and disease evolution, cancer heterogeneity and cancer genomics, antimicrobial resistance and hospital acquired infections, pandemic prevention and epidemic control.
- **AI for energy and environment** involves smart power grid design and grid resilience, fault tolerance, distribution networks and load distribution, contingency planning and optimization, renewables, grid security, environmental stewardship, air quality, water resource and critical infrastructure management, emergency planning – e.g., wild fires, flood, human influence and behavior modeling.

2) Investment in Foundational AI research and development

Foundational AI provides the building blocks of AI that can then be adapted and integrated to create use-inspired solutions. Foundational AI various core areas of AI including (but not limited to) machine learning, human-centric AI design, planning and control, vision and perception, robustness, explainability, interpretability and reasoning, and causal inference. Future research in foundational AI is needed in the following areas over the next 5 years:

- **Foundation models**, including large language models and world models, their design and limit studies, fidelity to real-world
- **Explainability and interpretability**, including reasoning tools, mechanistic interpretability, explainable AI, science-guided machine learning, low-complex analytical models to approximate large-parameter deep models
- **Scalable infrastructure for AI**, including energy-efficient data centers, microprocessor and chip design, high performance computing for training, chiplets and system-on-chip integration, energy harvesting devices and wearables, memory technology and processing-in-memory architectures, application specific integrated circuit design, parallel computing, low-cost training, reduction of model size without losing model power
- **Human-AI workflows**, including interactive workflows, human-robot collaboration, AI-assisted human workflow, human-guided AI workflow, AI-guided training and education, virtual and augmented reality environments for training and simulation, inclusive and accessible technology
- **Agentic AI** involving design of light-weight and reusable AI agents to tackle specific tasks including planning and scheduling, programming, data processing, and recommendation engines
- **Secure AI** including privacy preserving AI, federated learning and data privacy, computer and network security, cloud security, distributed systems security and resilience, frameworks for vulnerability assessment, cryptography, software and mobile security, web security.
- **AI test beds**, including experimental infrastructure for rapid prototyping and field testing in both controlled environments and close-to-real conditions, low-cost simulation environments and digital twins to pose what-if scenario

questions, and synthetic data fabrics to augment physical observations.

3) Workforce development and preparation

Adoption of AI relies on an AI-ready workforce across all sectors of science and technology. This takes training and raising AI awareness across the entire society, across all levels from K-12 through current workforce in industry. This holistic and comprehensive plan for workforce development and preparation will involve design of curricular materials and learning modalities that fit all levels, communication materials that are intended for public consumption and raising awareness, and collaborative and interdisciplinary tools intended for fostering research, innovation, and testing. While this requires innovation at all levels of education, new investment will be needed in the following dimensions to accelerate training and AI readiness of the U.S. workforce:

- **Responsible scaling plan:** Every area that is impacted by AI has some degree of automation with a potential to either augment the human performing a task or to potentially replace the human to perform the task. Therefore, new policy frameworks and long-term strategic planning are needed to chart a responsible scaling plan that would not only improve productivity and profitability but also empower the human productivity through creative partnerships at the human-AI interface, without jeopardizing the careers of the human workers who could be displaced due to AI. This will entail engaging across the entire spectrum of the workforce from the employers (owners, board of trustees) to the employees (managers, workers). Specifically, direct and trusted engagement with the workers on the field can not only ensure adequate preparation but also provide an opportunity to leverage their vast experiential knowledge to creating creative solutions for longer-term human-AI collaboration without fear of displacement. It could also help identify new opportunities where innovations are most needed with a high chance of community adoption.
- **Community-embedded training:** Traditional mechanisms of learning and training focus on structured institutions such as K-12 schools, 4-year colleges, and universities. However, accessibility to such formal institutions are often limited in rural communities across U.S. Studies have shown that

the best way to impart training at these rural communities is through an embedded model, where the training happens directly within the local communities. This could happen through associate degrees or through certifications and bootcamps, or simply through informal workshops and learning circles that aim to transfer expert knowledge to new learners. Providing a structure to this otherwise unstructured and unmeasurable training pathways is important. In addition, creating new learning pathways to on-board untrained workers into specialized positions will be needed.

- **Emphasis on K-12 education:** When it comes to AI, early exposure to the capabilities of AI and more broadly, computing and the power of data, are critical. This is most effectively carried out at the K-12 level, as students who are at the middle- and high-school levels make their career decisions. Students at this age are both curious and technologically adept, making them ideal targets for training on AI-driven technologies. However, safe and effective training kits are needed to help introduce AI and related technology (e.g. drones, robots, circuits) to this level, and to aid them in their exploration of science and technology. Programs that support cross-generational mentoring and peer-mentoring should be encouraged. In addition, efforts that train the teachers (e.g., train the trainer) need emphasis. Land-grant universities are best positioned to push the barriers in community education across all levels due to the programs that they already support in community outreach and extension.

4) Fostering academic-industry-public sector partnerships

The roles of the academic, private, and public sectors are complementing and synergistic. The academic sector innovates in AI research, leads discovery in science and engineering applications, and supplies an AI-ready workforce. The industry sector adopts, democratizes, and scales AI to fuel a job-creating economy, built around developing AI for enterprise-scale applications. The public sector envisions and solicits new development and innovation in social good applications and deploys them at scale within all layers of society, to create access and opportunity across the socioeconomic spectrum. This synergistic interplay between academia, public, and private sectors is symbiotic, as more jobs need more talent, research and innovation, and increased adoption and demand for scale.

5) Establishing research institutes and regional hubs

Establishing a network of research centers/institutes of excellence and regional hubs, housed and/or led by universities and in partnership with industry and public sectors, provide a systematic way to increase national prominence, and to catalyze innovation and accelerate workforce development. Much like the Manhattan project led to the establishment of national labs focused on energy, U.S. needs the next wave of centers of excellence across all U.S. regions. The National AI Research Institutes program, created in 2019, led to the establishment of 27 AI Institutes across the nation. They were each funded for a 5-year term. Leveraging the strong foundation and network laid by these institutes and capitalizing on their collective experience, time is now ripe for a long-term investment that would make these AI Institutes permanent and transform them into regional hubs of innovation in various areas of both foundational- and use-inspired AI. These institutes should emphasize research innovation coupled with translational research to practice and economic growth. Public-funded land-grant universities are ideally positioned to lead these institutes due to their deep-rooted connections to the region and communities they serve, investments from the state, and from the industry in the local region. Fortifying a set of permanent AI Institutes as regional hubs will serve as a long-lasting legacy.

6) Establishing global centers for international collaboration and knowledge sharing

Prominence in the global stage is critical to assert dominance in AI technology and its use. As the leading developer and exporter of AI technology and with a diversified economy across all sectors in the industry, U.S. is uniquely-placed to lead the AI revolution on the world stage. Key strategic partnerships with allies who also provide complementing technology and/or applications for U.S. technology, need to be identified, prioritized, and implemented. One effective way to implement such long-term strategic partnerships is through the establishment of a network of global centers that are supported by multiple countries across continental borders. These centers can pivot on partnerships that are bilateral, or on partnerships that involve multiple countries. Establishing such global centers can facilitate faculty and student exchange, knowledge and data sharing, and establish a reciprocal relationship and strong ties with the partnered countries, helping all countries including the U.S. collaboratively and cooperatively build and test AI technology in a trusted environment.